Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Functional and adaptive shape memory polymers for aerospace applications			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		**Chemicals**	
	Medium Category	Small Category	Medium Category	Small Category
	Aerospace Systems	Other Aerospace System Technologies	Polymeric Materials	Special Functional Materials Technologies
Concept of Proposed Technology	*Describe the core concept, main functions, and potential uses of the proposed technology* The proposed technology involves the fabrication of novel and functional components for aerospace applications using shape memory polymers (SMPs) as material plattform. SMPs are intelligent materials that are able to alter their shape upon applying an adequate stimulus. The change in shape can be triggered via solvent (such as water) [1,2], light [3,4] or via electric [5,6] or magnetic fields [7,8] but mostly heat [9–12]. During thermomechanical treatment („programming"),a temporary shape is imposed to obtain the SMP with its „one-way shape memory effect" (1W-SME). After programming this shape will be stable until an adequate stimulus is applied. Thereupon, the material recovers its original, permanent shape by the laws of entropy elasticty [13]			

	without any required additional external forces. This shape morphing behavior is repeatable once the material gets programmed again. Concerning the so called „two-way shape memory effect“ (2W-SME), the material is able to repeatedly change between two bistable shapes once programmed accordingly. These shapes can be attained by applying two consecutive stimuli, for single material systems in most cases heating and cooling [14–18]. Both, the 1W- and 2W-SME have thus the advantage to enable an autonomous material behavior and the functionality of a change in shape either passively due to changes in environmental conditions or actively via remote control. This is particularly beneficial for a vast range of applications in aerospace and -craft. An important representative of SMPs is phase segegrated thermoplastic polyurethane (TPU). TPUs are comprised of a soft segments (SS) and hard segments (HS). The SS are responsible for stabilizing the temporary shape after programming, while the HS enable the entropic driven shape recovery. The SS are built up by polyols (e.g. polyester-, polether or polycarbonate-diols) and the HS by diisocyanates and short-chained diols. By varying these molecular building blocks and their respective stoichiometry, the properties of the final TPU can be ultimatively adjusted in the synthesis. This allows for a precise customization of the material, from soft and elastic [19–21] to stiff and resistant [20–22], making TPUs attractive as a highly flexible and interesting material platform. Another advantage of TPUs is the possibility of thermal processing with conventional techniques, such as extrusion or injection molding, allowing for a later series production. In addition, TPU is highly eligible for additive manufacturing (AM), especially the fused filament fabrication (FFF) [23–25]. Next to material customization, this particularly opens up the possibility of free object and component design via AM. Recent studies also explored functional integrated additive manufacturing („4D printing“) [26,27]. This allows for a direct printing of stable stimuli-responsive objects, that are able to execute a shape morphing at a later state by applying a suitable trigger, such as heat [28,29]. The extent and direction of the SME can be controlled by choosing adequate printing parameters [28,30,31]. All these advantages, i.e. freedom of design of both material and (printed) components, together with shape memory behavior, offer the possibility of manufacturing functional, highly innovative and adapative elements, particularly suitable for aerospace appliactions. Potential applications include shape memory fixtures, which could be manufactured directly in space, requiring only a simple heat source (e.g., a heat gun) for activation and fixation. Further use cases include self-deployable habitats for lunar or Martian missions, as well as self-erecting structural elements for space stations, enabling simplified assembly and reduced reliance on complex mechanical systems. Further functional components include stimuli-responsive fasteners, hinges or dampers making the material platform highly suitable for applications in aerospace.
Rationale for Joint R&D Project	*Describe the need for this joint R&D project and why collaboration is required* The New Space sector is evolving rapidly, creating new opportunities in Earth observation, scientific missions, the Internet of Things (IoT), and broadband connectivity. Over the next nine years, the global small satellite market is projected to grow at a compound annual growth rate (CAGR) of 16.4%,

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	increasing from USD 6.9 billion to USD 30.6 billion [32]. In parallel, ambitious missions—including New Space stations, crewed Moon and Mars missions, and interplanetary exploration—are actively being prepared. These developments drive a strong demand for material and component innovation, particularly in the miniaturization of components and subsystems. Solutions are required that are not only lightweight and efficient, but also offer multifunctional capabilities, such as adaptive behavior, reduced mechanical complexity, and high reliability under extreme conditions. Many of these challenges can be addressed through the use of SMPs, leveraging their intrinsic shape memory properties for functional integration and adaptive system behavior. Yet, there are only limited materials available, matching these requirements for application in space. Those that are available (such as deployable solar panels, that can be folded using origami methods to save space during flight) on the other hand, exhibit only a fixed set of characteristics and are hardly adjustable in their properties. The Fraunhofer IAP can provide customized, responsive materials and components, tailored to the needs of the collaboration partner. Consequently, in a synergistic collaboration novel, functional materials and elements can emerge from the joint R&D project, benefiting both the Fraunhofer IAP and Korean SMEs.
Key Words	*Provide key terms that describe the proposed technology and its main features (min 3, max 5)* 1. Shape memory behavior 2. Autonomous and lightweight components for aerospace 3. Stimulus-responsive shape changes 4. Shape morphing 5. Self-healing

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (2)	**End of the Project**	Level (4)
R&D Objectives	*Describe the specific technologies you aim to develop through this R&D project* In the R&D project, functional and customized shape memory polymers are aimed to be developed and, together with the Korean partner and their vision, processed to functional components for aerospace applications. The project therefore results in following concrete technologies: - Novel shape memory polymers for use in aerospace - Novel designs and utilities for components in aerospace - Stimuli-responsive and functional items (such as fasteners, hinges or self-deployable structures, etc.) for diverse applications in aerospace			

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R&D Target Specifications	*Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters* Important performance specifications for exterior products are especially thermal stability, outgassing and ultraviolet resistance. A wide temperature span may occur, which the material must be able to withstand, ranging from -100°C (when the component is on the side facing away from the sun) and +100°C (when the material is on the side facing the sun). Especially, a general mechanical stability and resilience of the material needs to be ensured within this range of temperatures to avoid any loss in performance or material failure. Outgassing refers to the release of volatile substances (like additives, solvents or residual monomers) from a material, which is a crucial parameter for aerospace components as there is no ambient pressure inhibiting their release. These volatile molecules can condensate on cold surfaces such as optical lenses, cameras and sensors, resulting in incorrect measurements or a general loss of the overall performance of the element. Therefore, outgassing must be lower than 1 wt% according to NASA ASTM E595 [33]. UV resistance is also important, as there are no molecules present in a high vacuum to absorb UV radiation, meaning that the material is directly exposed to solar energy or rather UV light. Depending on the application and length or time span of the orbital mission, the material needs to withstand ≥1.000 ESH (Equivalent Sun Hours) to ≥20.000 ESH. When applying stimuli-responsive and shape morphing components, of course the material must exhibit a qualitative change in shape after programming and after applying a trigger. Measureable parameters are the “shape fixity ratio”, a relative measure in how far the polymer is able to hold the imposed temorary shape, and the “shape recovery ratio”, a relative measure in how far the poylmer is able to retain its original permanent shape after triggering the SME. These values are best to be as high as possible, but ≥90% for the shape fixity ratio and ≥70% for the shape recovery ratio.
R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives* 1) As a first work package, the exact application and need of the Korean SME partner will be elaborated and evaluated. This will be translated into specification sheet summarizing basic requirements for a suitable functional material, i.e. thermal properties such as melting and/or glass transition temperature, degradation temperature, etc., mechanical properties such as Young’s modulus, stress and strain at break, etc., and the shape memory behavior such trigger, switiching temperature regime, fixitation and recovery ratio, etc. Milestone: Specification sheet for a shape memory TPU is set. Timeline: Month 1 - 3 2) Using the specification sheet, different TPU formulations will be theoretically developed, including their synthesis in lab scale (max. 250 g per batch) in several iterations. The prepolymer method will be used to synthesize polyester-, polyether- or polycarbonate-based TPUs with shape memory properties. The materials will be characterized to derive structure-property-relationships. This knowledge aids to finally synthesize an SMP meeting the criteria from work package 1. If one (or more) formulations were identified as suitable polymers, the synthesis will be scaled to kilogram scale

for further processing.
Milestone: Succesful synthesis and scaling of a suitable TPU.
Timeline: Month 3 - 12

3) The synthesized polymers from work package 2 will be characterized thoroughfully. In the beginning, the mechanical properties will be anaylzed using a universal testing machine to determine the Young's modulus and material failure. Differential scanning calorimetry is used to analyze the thermal behavior, i.e. phase transitions of the material and thermogravimetric analysis for decompostion temperature, thus provoiding information regarding thermal stability. Cyclic thermo-mechanical measurements are conducted to evaluate the shape memory properties (shape fixity and shape recovery ratio). These results will be used to draw conclusions about the structure-property-relationships in order to finally produce a suitable SMP in accordance with Work Package 1. If a suitable material is found, application-oriented analysis will be conducted such as dynamic mechanical analysis to screen the mechanical stability over a temperature range and/or to examine possible creep, outgassing tests as well as UV resistance. According to the results, minor adjustments in the material formula can be made and especially the use of different additives in the synthesis will be evaluated for material performance.
Milestone: A suitable and applicable TPU with shape memory behavior is evaluated.
Timeline: Month 5 - 14

4) The scaled and suitable material(s), will be further processed to filaments for later rapid prototyping. Therefore, the SMP will be granulated and extruded afterwards. The aim is to produce usable filaments of high quality that are able to be printed using FFF. Therefore, the optimal extrusion parameters will be investigated (such as temperature of heating zones, screw speed, torque, etc.) and adapted to the material.
Milestone: A filament with diameter 2,85mm ± 0,1mm or 1,75mm ± 0,1mm is developed successfully.
Timeline: Month 14 - 18

5) The filaments from work package 4 are used for AM (normal or function integrated). Before functional components are printed, first the printing paramters will be tested. These include printing platform temperature, print speed, nozzle temperature and the influence of printing pattern as these parameters affect the quality and especially mechanical resilience of the 3D printed objects. Thus, the parameters will be adapated and evaluated for the novel material using simple geometries, like cuboic or cylindrical samples to identify the optimum printing conditions. To identify suitable parameters, the manufactured printed objects are analyzed microscopically for defects.
Milestone: The optimum parameters for the material are identified and first 3D-printed objects of good structural resolution have been obtained.
Timeline: Month 18 - 22

6) In parallel, CAD models will be designed in strong collaboration with the project partner. The

	models combine the envisioned application, stimulus-responsive functionality and evaluated material properties. The interplay between these factors is taken into account during modeling different designs and their synergistic effect that is utilized to produce functional components for later use in aerospace. The designs will be printed by applying the evaluated printing parameters from work package 5. Similar to the iterative material synthesis, final demonstrators will be developed in printing cycles including remodelling of the designs and printing adjustments, such as object infill, wall thickness, utilizing supporting structures, etc. in high quality. Milestone: Succesfull additive manufacturing of functional components. Timeline: Month 18 - 26 7) The components from work package 6 will then be tested and evaluated in lab environmet. Especially their stimuli-responsivity – the shape memory behavior – is analyzed and evaluated, whether the extent of change in shape is sufficient and whether it matches the envisioned funcionality of the component regarding the application. Points to consider are, for example, if the mechanical properties are sufficient, if the programming route is suitable, if there is a secure and accurate form fit after triggering the SME, if the change in length is sufficient and if the geometry corresponds to the theoretically required form. With this knowledge, both the design and the printing patterns can be adjusted in order to finally print suitable and usable demonstrators. Milestone: Component was tested to be functional and suitable for the envisioned aerospace application. Timeline: Month 22 - 30 8) Filaments from work package 4, CAD designs from work package 6 and/or demonstrators from work package 7 will be provided to the Korean SME considering their exact needs and expectation. It is then aimed to do case studies and field tests by the Korean partner, best under environmental (artificial space) conditions. The functional components will be integrated or assembled to the envisioned device and analyzed, if the SME aids the constructional element and thus enhances the functionality of the device. The result will be evaluated if a potential application in space is highly imaginable. Milestone: The component was tested (under environmental conditions) in devices as first application demonstrators. Timeline: Month 30 - 36 9) Finally, the utilization of the results is considered and how to proceed. If the technology is not yet mature, follow-up projects should be initiated together with the partner in order to bring the technology to market maturity. If the results are more than promising, a direct market transfer can take place, e.g. via joint patenting and licensing for, in the best case, an immediate use by the partner SME.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.* The Fraunhofer IAP will take care of material development, material analysis as well as processing and thus contribute with their expertise in customizing and programming of SMPs. Specifically, a novel material formulation will be developped according to the given requirements by the partner. Furthermore, the Fraunhofer IAP will contribute with their know-how in filament extrusion and (functional integrated) fused filament fabrication. The Korean partner specifies their functional application, which is translated to a specific CAD design in collaboration, connecting the expertise of shape memory behavior from the Fraunhofer IAP and the construction of aerospace components from the Korean partner. The filaments and/or 3D printed components, aligning the envisioned aerospace application, will then be provided to the Korean SME for testing and characterization.
Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.* The Korean SME will provide their specific needs to their intended aerospace components, which will be translated into a material specification sheet. This will be provided to Fraunhofer IAP for material synthesis. Additionally, they communicate their vision for design and functionality, i.e. specific models and shape morphing behavior, which will be taken into account during designing different CAD elements by both, Fraunhofer IAP and the Korean partner in collaboration to combine their expertises synergistically. Lastly, the SME partner will contribute with their expertise in the field of aerospace systems and owe the task to test the shape memory polymer and/or component in application relevant scenarios or similar conditions for applicability and suitability of the material properties, component designs and shape memory functionality.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.* The expected outcomes are technology demonstrators for aerospace and space applications. These demonstrators are devices specifically designed, manufactured and tested for operation in space-relevant environments. The key innovation lies in the integration of functional, stimuli-responsive components made from SMPs as an integral part of the demonstrators. From the technological perspective, such autarkic components are novel and therefore highly innovative with significant technological advancements for the aerospace sector. Hence, the functional components as well as the adjustable material platform by itself shows a strong commercialization potential, following a further maturation and qualification of the technology to industrial prototypes. A possible commercial launch lies in the licensing of joint IP and/or, at best, directly by the project partner itself, enabling a

	faster market transfer. Besides, the developed concepts for the stimuli-responsive SMP components are higly versatile and can be transferred to wide range of aerospace applications and can be adapted for the integration into other devices and systems. Particularly, this does not necessarily have to take place in the context of aerospace but can also be transferred to a broader range of industrial sectors that demands innovative, novel and functional components, such as the aircraft, marine, or automotive industry.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization